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Energy Supply Resilience and Industrial Continuity Under a Strait of Hormuz Blockade

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Abstract

A blockade or severe disruption in the Strait of Hormuz would test energy supply resilience by reducing crude oil and LNG availability and by raising routing, freight, insurance, port-handling, warehousing, and transport-support costs. This paper develops a short-run multi-regional input–output stress test to assess where such an energy-route shock enters the production system, how reserves and inventories reduce pass-through, which cross-border links carry residual costs, and where final demand absorbs them. Using the OECD ICIO 2025 edition for 2022, we map the shock to oil and gas extraction, refining, utilities, transport, and transport-support sectors, with an additional premium for major Gulf energy exporters. We propagate the shock for seven input–output rounds under inventory damping. First-round exposure and later-round burden do not coincide, as energy-intensive materials, aviation services, chemicals, minerals, metals, electronics, and machinery face higher downstream costs through material and logistics purchases. With 30% inventory absorption, the upstream energy shock needed for downstream manufacturing to reach a 10% added-cost threshold rises from 73.6% to 85.3%. The results support targeted reserve release, coordinated rerouting, port-logistics priority, inventory management around high-value links, and continuity protection for vulnerable sectors.

Keywords: energy security; Strait of Hormuz; energy supply disruption; maritime chokepoints; strategic petroleum reserves; supply-chain resilience; multi-regional input–output analysis; industrial continuity

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1. Introduction

The Strait of Hormuz is usually measured as a transit corridor for crude oil and LNG. Energy agencies identify it as one of the main passages for oil and LNG trade [1–3], and recent market reports emphasize its continuing role in Middle East energy supply [4,5]. A blockade or near-closure would therefore not be only an import-dependence event. Crude and LNG cargoes would need to be replaced, rerouted, stored, or rationed before refinery runs, electricity and gas services, and energy-intensive production began to adjust. Tanker routing, freight rates, war-risk insurance, port handling, warehousing, and transport-support services would also change the delivered cost of energy and energy-related inputs. The first accounting entries would appear around Gulf energy export nodes, refining systems, utilities, maritime transport, and related support services. Later entries would be less visible in a trade-flow dashboard, because industrial users buy both energy and intermediate goods whose own costs already contain energy, logistics, and transport-support inputs.

This paper asks three energy-security questions. Which energy and logistics sectors receive the initial cost pressure under a severe Hormuz disruption? After reserve release and inventory absorption are allowed, which countries, sectors, and cross-border input links continue to carry residual costs? Which policy levers—reserve release, supply diversification, rerouting, logistics coordination, inventory management, and industrial-continuity protection—are most closely connected to those transmission points?

Energy-security research begins from availability, affordability, strategic stocks, emergency response, and continuity of delivered energy services [6]. Studies of oil-import and supply-chain security examine import concentration, strategic reserves, and supply-chain exposure [7,8]. Chokepoint and LNG-route studies add the maritime dimension of route dependence and spare capacity [9,10]. Work on oil-supply disruption, system resilience, and market shocks shows why reserves, spare routes, and crisis instruments matter in the short run [11–13]; related studies of oil-price and geopolitical shocks clarify that observed price movements mix supply, demand, and policy responses [14–16]. These literatures describe the first stage of the problem with useful precision. They are less explicit about where any residual cost pressure goes after reserves, rerouting, inventories, and contracts have absorbed only part of the shock.

Production-network research addresses that second stage. It shows that aggregate outcomes depend on network position, input specificity, asymmetric dependence, and nonlinear propagation [17–19]. Other work formalizes macroeconomic amplification through input–output linkages and production-network structure [20–22]. Disaster and firm-level studies show how interruptions can move through supplier–customer relations rather than staying near the original disturbance [23–25]. Pandemic supply-chain studies make a similar point for broad production shutdowns [26,27], while input–output restart models emphasize sectoral constraints and demand propagation [28,29]. What is still less common is a calculation that carries the same maritime-energy shock from source sectors, through intermediate cost-transfer accounts, to final-demand destinations. For a severe Hormuz disruption, this means treating route constraints, reserves, inventories, sticky procurement relations, and destination-country demand as parts of one accounting exercise.

This study develops a dynamic multi-regional input–output (MRIO) stress test for that short-run object. The empirical backbone is the 2022 country-sector transaction matrix in the 2025 OECD Inter-Country Input–Output release [30]. The model maps a Strait of Hormuz blockade into a node-level shock vector for oil and gas extraction, refining, utilities, transport, and transport-support activities, with an additional route-disruption premium for major Gulf energy exporters. Added costs are then transmitted round by round through the technical-coefficient matrix, while inventory damping reduces the share of each round's cost pressure that is passed forward. The exercise is an accounting stress test rather than an oil-price forecast: it reports where the 2022 production structure would carry a cost increase, not how traders, fleets, contracts, or macroeconomic demand would adjust in real time.

The contribution is to connect an energy-route disruption to the country-sector accounts, cross-border input links, inventory buffers, and final-demand destinations that matter for energy supply resilience. The framework separates the point where the shock begins from the places where later costs are recorded, and it links the resulting transmission points to policy instruments that can still operate during the short-run adjustment window: reserve release, substitute routing, port and freight coordination, inventory management, and industrial-continuity protection.

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2. Data, Disruption Scenario, and Resilience Stress-Test Method

2.1. Data Processing

The analysis uses the 2025 OECD ICIO release for the 2022 benchmark year as its main data source [30]. The method is a short-run energy supply resilience stress test: it identifies where an energy-route disruption enters the production system, how reserve and inventory buffers attenuate pass-through, which country-sector links carry residual costs, and where the burden is finally absorbed. The benchmark year is used because it provides a complete post-pandemic country-sector table with the China and Mexico subregional nodes needed for link extraction before national reaggregation. After removing total accounts, tax accounts, value added, final-demand accounts, and other non-production entries, the working sample contains 4200 production nodes covering 84 country or region codes and 50 sectors. Let C denote countries or regions, R sectors, and $N = C \times R$ production nodes. The final-demand accounts are reassembled by destination country; for country-level reporting, CN1 and CN2 are merged into CHN, and MX1 and MX2 into MEX. Cross-border link extraction retains the original subregional nodes so that high-value edges are not averaged away. The annual MRIO table cannot reproduce weekly vessel movements, daily freight rates, insurance premia, or market-clearing policy effects. The test asks where the existing procurement structure carries added cost once the entry shock is imposed. Table 1 summarizes the workflow, and Table 2 defines the main symbols.

Table 1. Workflow from energy-route scenario to resilience indicators.

Step	Input	Core Operation	Output
1	2025 OECD ICIO release for 2022	Remove total accounts, taxes, value added, and other non-production items. Retain only country-sector production nodes and intermediate-input flows.	Production-node set N, intermediate-input matrix Z, gross-output vector x
2	Original final-demand accounts	Reaggregate household consumption, government consumption, capital formation, and related accounts by destination country.	Final-demand matrix F
3	Energy-system source classification and scenario setting	Define the shock-source sector set R_{src}, the pure downstream reporting set D, and the Gulf export node set G.	Energy-entry boundary and downstream reporting rules
4	Baseline disruption parameters	Translate route disruption, substitute capacity, freight escalation, and Gulf export premia into the shock vector s.	Initial energy-logistics shock vector s
5	Reserve and inventory information	Set the inventory absorption rate γ and the damping matrix Γ as reduced-form mitigation capacity.	Node-level attenuation mechanism for residual pass-through
6	Technical-coefficient matrix A and shock vector s	Propagate residual cost pressure round by round and obtain $p^{(h)}$ and cumulative propagated cost $P^{(H)}$.	Direct exposure, subsequent cascades, and cumulative resilience stress
7	Node-level cumulative propagated cost	Aggregate to countries and sectors using output weights. Extract high-value propagated cost flows with $\mathrm{diag}(P^{(H)})Z$.	Policy-relevant country, sector, and critical-link indicators
8	Node-level cumulative propagated cost and final-demand matrix F	Compute propagated final-demand incidence and the resilience threshold surface $\Phi(\alpha, \gamma; H)$.	Inventory-boundary and final-market incidence results

Table 2. Key symbols and definitions.

Symbol	Definition
N	Set of cleaned production nodes, where each node is a country–sector combination
C	Set of countries or regions, with element c denoting one economy
R	Set of sectors, with element r denoting one sector
$c(i)$	Economy to which node i belongs
$r(i)$	Sector to which node i belongs
$Z = (z_{ij})$	Intermediate-input matrix, where z_{ij} is the intermediate flow supplied by upstream node i to downstream node j
$x = (x_1, \ldots, x_n)^\top$	Gross-output vector, where x_j is the gross output of node j
$A = (a_{ij})$	Technical-coefficient matrix, where $a_{ij} = z_{ij}/x_j$ measures the direct dependence of node j on node i per unit of output
$F = (f_{ik})$	Final-demand matrix, where f_{ik} is the final demand for node i absorbed in destination country k
$s = (s_i)$	Exogenous shock vector describing the initial energy-route disruption cost imposed on source nodes under the Strait of Hormuz scenario
E, U, T	Sets of energy extraction and refining sectors, utilities sectors, and transport and transport-support sectors, respectively,
G	Set of economies directly constrained by Gulf export-route disruption
R_{src}	Shock-source sector set, with $R_{\text{src}} = E \cup U \cup T$
D	Pure downstream reporting-node set, defined by excluding the shock-source sectors from the reporting sample to isolate industrial-continuity effects
$\mathcal{D}_c$	Pure downstream node subset belonging to country c
$\mathcal{D}_r$	Pure downstream node subset belonging to sector r across all economies
C^{inv}	Set of economies with inventory buffering capacity
$\Gamma = \text{diag}(\gamma_i)$	Reserve and inventory damping matrix, where γ_i is the share of shock pressure temporarily absorbed by the economy of node i
g	Effective supply-gap rate that remains unfilled after route disruption and substitute pipelines are both taken into account
$p^{(h)}$	Incremental cost-pressure vector in round h
$P^{(H)}$	Cumulative propagated cost vector through round H; it excludes the source-node's own shock s except when that shock returns through input–output feedback
H	Length of the short-run crisis-response propagation window in the stress test
$\Delta V_c^{(h)}$	Output-weighted incremental cost pressure absorbed by country c in round h
$E_i^{(1)}$	First-round direct exposure of node i after the initial shock is passed to direct buyers
$C_i^{(H)}$	Subsequent cascade pressure absorbed by node i from round 2 to round H
$T_i^{(H)}$	Total cost pressure accumulated by node i over the reporting window
$E_c^{(1)}$	Output-weighted first-round direct exposure of country c
$T_c^{(H)}$	Output-weighted total cost pressure accumulated by country c over the reporting window
$E_r^{(1)}$	Output-weighted first-round direct exposure of sector r
$C_c^{(H)}$	Subsequent network-cascade pressure absorbed by country c from round 2 to round H
$C_r^{(H)}$	Subsequent network-cascade pressure absorbed by sector r from round 2 to round H
$T_r^{(H)}$	Output-weighted total cost pressure accumulated by sector r over the reporting window
$M^{(H)}$	Propagated cost-transfer matrix, where $m_{ij}^{(H)}$ is the absolute added cost transferred from node i to node j after the entry shock has moved through the network
$L^{(H)}$	Propagated final-demand incidence matrix, where $l_{ik}^{(H)}$ is the network-transmitted cost generated by node i and absorbed by destination country k
Y_k^F	Nominal destination final demand of country k computed from the cleaned production-by-final-demand block
$R_k^{(H)}$	Propagated final-demand burden of country k as a share of Y_k^F
$\Phi(\alpha, \gamma; H)$	Downstream manufacturing resilience indicator, defined as the average cost increment after H rounds under upstream-energy shock level α and inventory absorption rate γ
$\alpha^*(\gamma; \tau)$	Critical upstream-energy shock level associated with threshold τ under inventory absorption rate γ

2.2. Intermediate-Input Propagation and Final-Demand Incidence Matrices

The stress test separates intermediate-input relay structure from final-demand incidence, following standard input–output logic [31,32]. The ICIO production block gives the intermediate-input matrix $Z = (z_{ij}) \in \mathbb{R}_+^{n\times n}$, where z_{ij} is the monetary flow from upstream node i to downstream node j and $n = |N|$. With gross output $x = (x_1, \dots, x_n)^\top$, the technical-coefficient matrix $A = (a_{ij})$ is

$$a_{ij} = \frac{z_{ij}}{x_j}, \qquad x_j > 0. \tag{1}$$

Thus, a_{ij} is the direct input requirement from node i per unit of output in node j. Zero-output nodes are removed before this calculation.

The price side is normalized to the benchmark ICIO price system. Each node is treated as having benchmark unit price $q_i^0 = 1$, and s_i, $p_i^{(h)}$, and $P_i^{(H)}$ are interpreted as relative added-cost wedges against that normalized 2022 nominal base. This normalization is standard for a price-side input–output stress test: it identifies cost incidence under fixed benchmark input shares, not a new market-clearing price vector or a decomposition of price and quantity responses.

Final incidence is identified separately. Household consumption, government consumption, capital formation, inventory changes, and related final-demand accounts are reaggregated by destination country to form $F = (f_{ik}) \in \mathbb{R}^{n\times|C|}$. If F_k is the set of final-demand accounts flowing to destination country k, then

$$f_{ik} = \sum_{m\in F_k} y_{im}, \tag{2}$$

where y_{im} is the supply of node i to original final-demand account m. The real-valued range is retained because inventory-change entries can be negative. Reaggregation by destination country is needed because the analysis asks where cumulative costs are finally absorbed by national markets.

A determines the intermediate relay structure. F determines the final markets where cumulative costs are absorbed. Keeping them separate avoids mixing transmission paths with final incidence.

2.3. Energy-Route Disruption Scenario and Shock-Source Mapping

Scenario construction translates the energy-security disturbance into the source nodes that first experience pressure. The exogenous disturbance is a node-level relative cost increment vector $s = (s_i)_{i\in N}$. Let E denote energy extraction and refining sectors, U utilities, T transport and transport-support sectors, and G the economies receiving the additional Gulf export-route premium. The vector combines a broad delivered-cost increase for energy, utilities, and transport-related accounts with a Gulf-specific route premium η imposed only on energy nodes in the major route-constrained exporters. Supplementary Table S2 reports the exact ICIO sector codes and the Gulf export node set. In the working table, $G = \{\mathrm{SAU}, \mathrm{IRN}, \mathrm{IRQ}, \mathrm{UAE}, \mathrm{KWT}, \mathrm{QAT}, \mathrm{OMN}\}$. Bahrain is retained for country reporting but is not included in G; a high BRN ranking is therefore interpreted as a network-position result, not as a direct Gulf-node premium. For node i,

$$s_i = \begin{cases} \alpha_E + \eta, & c(i) \in G,\ r(i) \in E, \\ \alpha_E, & c(i) \notin G,\ r(i) \in E, \\ \alpha_U, & r(i) \in U, \\ \alpha_T, & r(i) \in T, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

In Equation (3), $c(i)$ identifies the economy and $r(i)$ the sector. Parameters α_E, α_U, α_T, and η are relative cost increments for energy and refining, utilities, transport and logistics, and Gulf export-route exposure, respectively. For example, $\alpha_E = 0.20$ means a 20% increase relative to baseline. This is a cost-incidence stress vector, not a physical rationing model. Nodes outside E, U, or T do not receive an exogenous shock initially, but they can still absorb cost pressure later. Resource contraction, transport disruption, and utility-cost escalation enter through related channels [8,9,11]; route and reserve studies motivate the inclusion of maritime, spare-capacity, and stock-release terms [10,12,13]. Utilities and transport-support activities are included because energy costs move quickly into refining, chemicals, metals, port handling, warehousing, and integrated logistics services. Air transport is not assigned an exogenous shock in the baseline; if it ranks high later, that result comes from fuel-intensive inputs and network position.

These parameters are anchored to route constraints, substitute pipeline capacity, and freight-rate multipliers. Let $\mathrm{Cap}^{\mathrm{Hormuz}}$ denote the normal traffic capacity of the Strait of Hormuz, let θ_t denote the proportion of capacity disrupted, let $\mathrm{Cap}^{\mathrm{pipe}}_{\mathrm{SA}}$ and $\mathrm{Cap}^{\mathrm{pipe}}_{\mathrm{UAE}}$ denote the substitute pipeline capacities available to Saudi Arabia and the United Arab Emirates, respectively, and let m_f denote the freight-rate multiplier relative to normal conditions. The effective supply-gap rate that remains unfilled after disruption and partial substitution is written as

$$g = \frac{\max\left(\theta_t \mathrm{Cap}^{\mathrm{Hormuz}} - \mathrm{Cap}^{\mathrm{pipe}}_{\mathrm{SA}} - \mathrm{Cap}^{\mathrm{pipe}}_{\mathrm{UAE}},\ 0\right)}{\mathrm{Cap}^{\mathrm{Hormuz}}}. \quad (4)$$

On that basis,

$$\alpha_E = \phi_E g, \qquad \alpha_U = \phi_U \alpha_E, \qquad \alpha_T = \phi_T (m_f - 1), \qquad \eta = \phi_H g. \quad (5)$$

Here, $g \in [0, 1]$ is the supply gap left after route disruption and pipeline substitution. The coefficients map that gap into upstream energy, utilities, transport-support, and Gulf-node premium terms. The gap term is an oil-flow proxy, not a full oil-and-LNG rerouting model. Saudi and UAE bypass capacities are scenario allowances within the range of public estimates, not guaranteed immediately available spare capacity. LNG disruption enters through the transport, utilities, and Gulf-node premium terms rather than through a separate pipeline-offset equation.

The coefficients ϕ_E, ϕ_U, ϕ_T, and ϕ_H are calibration coefficients used to translate physical route disruption into the baseline stress vector. They are not estimated elasticities. We choose them so that the baseline scenario corresponds to a 20% upstream energy shock, a 15% utilities shock, a 25% transport/logistics shock, and a 10% Gulf-node premium. For the threshold analysis, α denotes the upstream-energy shock level, so the baseline is $\alpha = 20\%$. Other components keep the baseline composition, with $\alpha_U = 0.75\alpha$, $\alpha_T = 1.25\alpha$, and $\eta = 0.50\alpha$. Robustness to inventory assumptions is reported separately. Table 3 reports the baseline parameter values used in the stress test, and Supplementary Table S1 reports the underlying physical inputs, scaling coefficients, and implied baseline shocks in one place.

For risk reporting, source sectors must be separated from pure downstream recipients. Let $R_{\text{src}} = E \cup U \cup T$ denote the set of shock-source sectors. The pure downstream receiving-node set is then defined as

$$D = \{\, i \in N : r(i) \notin R_{\text{src}} \,\}. \tag{6}$$

Equation (6) changes the reporting sample rather than the propagation mechanism. Source sectors still enter the model through vector s and transmit pressure to downstream nodes. They are excluded from the pure downstream reporting set when the analysis reports country burdens, sectoral subsequent-burden rankings, and the manufacturing threshold surface. The cross-sector transfer matrix, however, is still constructed from the full propagated edge matrix so that source and transport-support sectors can appear as transmitting sectors. Without this distinction, direct source shocks would be mixed mechanically with later-round downstream burden.

Table 3 reports the baseline parameter values. The baseline is a central stress scenario used for comparison, not a forecast of the most likely future path. The country rankings, sector rankings, and threshold boundaries reported below show how structural position changes outcomes under this setting. The values 20%, 25%, and 10% are calibrated scenario inputs rather than uniquely correct market numbers. They are more suitable for comparison and screening than for direct extrapolation into daily price forecasts.

Table 3. Baseline scenario and parameter settings.

Module	Baseline Setting	Explanation
Real-world anchor	Severe disruption of Strait of Hormuz navigation under Middle East escalation risk	Uses IEA and EIA information on chokepoint traffic and inventories as the empirical background; it is not an ex-post reconstruction of one exact event path
Upstream energy and refining shock α_E	20%	Applied to oil and gas extraction, refining, and related upstream energy nodes
Utilities shock α_U	15%	Reflects higher costs of electricity, gas, and public energy services
Transport and transport-support shock α_T	25%	Captures higher shipping, rerouting, insurance, and handling-support costs
Premium on Gulf export nodes η	+10%	Additional route-disruption premium imposed on major export nodes
Propagation rounds H	7 rounds	Used to identify first-round exposure, subsequent amplification, and cumulative burden
Inventory-buffer country pool	GER, FRA, GBR, IND, JPN, KOR, USA	Baseline setting based on OECD commercial inventories and the U.S. SPR; GER denotes Germany, which is coded as DEU in the original ICIO data
Baseline inventory absorption rate γ	30%	Indicates that inventory-holding economies can absorb 30% of incremental cost pressure locally
Threshold grid	$\alpha \in [10\%, 100\%]$, $\gamma \in [0, 50\%]$	α is the upstream-energy shock level; $\alpha_U = 0.75\alpha$, $\alpha_T = 1.25\alpha$, and $\eta = 0.50\alpha$ preserve the baseline composition used to construct the threshold surface

2.4. Reserve and Inventory Damping in the Short-Run Propagation Window

In the present framework, inventories and strategic reserves are treated as short-run mitigation capacity, not as a full replacement for disrupted supply. They are modeled as the share of cost pressure that can be absorbed before each round of onward transmission. This reduced-form treatment is intended to capture the damping effect of public stock release, commercial inventories, and firm-level buffers without modeling the release order, ownership structure, or market-clearing price response of each stock category. Let C^{inv}

denote the set of economies with inventory-buffering capacity. Let $\gamma_i \in [0,1]$ denote the inventory absorption rate of node i, and construct the diagonal matrix $\Gamma = \mathrm{diag}(\gamma_1, \dots, \gamma_n)$. Under the baseline setting, if $c(i) \in C^{\mathrm{inv}}$ then $\gamma_i = \gamma$, otherwise $\gamma_i = 0$, namely

$$\gamma_i = \gamma \cdot \mathbb{I}\{\, c(i) \in C^{\mathrm{inv}} \,\}. \tag{7}$$

In Equation (7), the common parameter γ is the uniform share of pressure that buffer economies can absorb locally. The buffer set is GER, FRA, GBR, IND, JPN, KOR, and USA, selected from internationally comparable commercial-inventory information, IEA emergency-stock information, U.S. Strategic Petroleum Reserve data, and evidence on substitution between public and private oil storage [33–36]. The notation GER is used in tables for readability, while the underlying ICIO code for Germany is DEU. China is not included in the baseline pool because the rule is restricted to directly comparable stock series plus explicit U.S. SPR information; this is a data-consistency choice, not a claim that China lacks inventories. The treatment remains reduced-form and does not distinguish release order, product composition, ownership, or substitute-procurement timing. Related work shows that inventories and adaptive behavior change the timing profile of losses [37,38]. Supplementary Table S3 extends the baseline with an expanded buffer-country pool that adds China and a sector-specific damping rule.

Strategic reserves and commercial inventories enter at different points. The shock vector defines the pressure remaining after assumed bypass capacity and source-side mitigation. The damping matrix then represents the share of incremental pressure that inventory-holding economies absorb before passing costs to downstream buyers. The damping parameter is therefore a reduced-form stock buffer, not a release schedule for public or commercial stocks.

Given this setup, incremental costs enter the network round by round. Round 1 records the transmission of source-node cost increases to direct buyers through the existing procurement structure. From Round 2 onward, part of the previous round's pressure is absorbed locally by inventories and the remainder is passed on through input linkages. Let $p^{(h)} \in \mathbb{R}^n$ denote the incremental cost-pressure vector in round h, where $p_i^{(h)}$ is the added pressure absorbed by node i in round h. Let H denote the propagation window. Then

$$p^{(1)} = A^\top s, \tag{8}$$

$$p^{(h)} = A^\top (I - \Gamma) p^{(h-1)}, \qquad h = 2, \dots, H, \tag{9}$$

$$P^{(H)} = \sum_{h=1}^{H} p^{(h)} = \sum_{h=0}^{H-1} \left[A^\top (I - \Gamma)\right]^h A^\top s. \tag{10}$$

In Equation (8), $A^\top$ allocates the source-node cost shock to direct buyers according to input shares. In Equation (9), $(I - \Gamma)p^{(h-1)}$ is the residual pressure not absorbed locally, and $A^\top$ transmits it to the next round of buyers. Equation (10) gives the cumulative propagated cost vector $P^{(H)}$. The round index is an input–output relay layer, not a calendar unit. The reported country, link, and final-demand matrices use $P^{(H)}$, not $s + P^{(H)}$, so they measure downstream incidence after the entry shock has moved through the MRIO network. A finite window with $H = 7$ is used to capture the near-term propagation depth relevant for reserve use, logistics coordination, and industrial continuity. Supplementary Figure S1 shows that the curves flatten after about seven to eight rounds.

To distinguish entry exposure, subsequent amplification, and the total burden within the window, the following node-level metrics are defined

$$E_i^{(1)} = p_i^{(1)}, \qquad C_i^{(H)} = \sum_{h=2}^{H} p_i^{(h)}, \qquad T_i^{(H)} = E_i^{(1)} + C_i^{(H)} = P_i^{(H)}. \tag{11}$$

Here, $E_i^{(1)}$ is first-round direct exposure, $C_i^{(H)}$ is subsequent cascade pressure, and $T_i^{(H)}$ is total accumulated pressure. Reporting them separately avoids mixing a large entry shock with prolonged later-round accumulation.

2.5. Aggregation, Critical-Link Extraction, and Structural Indicators

The stress-test outputs are reported with output-weighted aggregation so that very small nodes do not dominate country or sector rankings. Country burdens and sectoral subsequent-burden rankings use the pure downstream node set D. Source and transport-support sectors still appear in the cross-sector transfer matrix when they transmit propagated costs. For country c, let $\mathcal{D}_c = \{\, i \in D : c(i) = c \,\}$ denote its pure downstream node subset. Country internal weights are

$$\omega_i^c = \frac{x_i}{\sum_{j \in \mathcal{D}_c} x_j}, \qquad i \in \mathcal{D}_c, \tag{12}$$

where the denominator is total gross output of pure downstream nodes in country c. The country-level incremental burden in round h is

$$\Delta V_c^{(h)} = \sum_{i \in \mathcal{D}_c} \omega_i^c p_i^{(h)}, \tag{13}$$

$$E_c^{(1)} = \Delta V_c^{(1)}, \qquad C_c^{(H)} = \sum_{h=2}^{H} \Delta V_c^{(h)}, \qquad T_c^{(H)} = \sum_{h=1}^{H} \Delta V_c^{(h)}. \tag{14}$$

Here, $E_c^{(1)}$, $C_c^{(H)}$, and $T_c^{(H)}$ are first-round exposure, subsequent cascade pressure, and total burden. Sector-level aggregation follows the same logic and is evaluated on the pure downstream node set so that rankings are not mechanically dominated by source sectors. Let $\mathcal{D}_r = \{\, i \in D : r(i) = r \,\}$ denote the downstream node subset of sector r. Then

$$\omega_i^r = \frac{x_i}{\sum_{j \in \mathcal{D}_r} x_j}, \qquad \Delta V_r^{(h)} = \sum_{i \in \mathcal{D}_r} \omega_i^r p_i^{(h)}, \tag{15}$$

$$E_r^{(1)} = \Delta V_r^{(1)}, \qquad C_r^{(H)} = \sum_{h=2}^{H} \Delta V_r^{(h)}, \qquad T_r^{(H)} = \sum_{h=1}^{H} \Delta V_r^{(h)}. \tag{16}$$

Equations (15) and (16) define sector first-round exposure, subsequent cascades, and total burden. The aggregated indicators can be read as average added cost per unit of output at the corresponding reporting level.

Gross output is used as the aggregation weight because the outcome being reported is a production-continuity cost-pressure indicator, not GDP loss or welfare loss. The propagation equations and the technical coefficients are defined on gross-output input shares, so output weighting keeps the country and sector averages on the same price-normalized production basis. This choice gives more weight to sectors with large intermediate purchases, including refining, chemicals, metals, transport, and equipment supply chains. That is appropriate for tracing pass-through through production accounts, but it also means the aggregated indicators should not be interpreted as value-added losses. The usual input–output double-counting concern remains: intermediate-intensive sectors are represented in proportion to their gross production scale. We therefore use the output-weighted values as structural cost-incidence rankings and discuss this limitation separately.

Country and sector averages still do not reveal the specific transaction edges along which costs continue to move. The analysis remaps node-level cumulative propagated cost onto the intermediate-input flows and defines the propagated cost-transfer matrix

$$M^{(H)} = \mathrm{diag}(P^{(H)})Z, \tag{17}$$

whose elements are

$$m_{ij}^{(H)} = P_i^{(H)} z_{ij}. \tag{18}$$

In Equation (18), $P_i^{(H)}$ is the cumulative propagated relative cost of node i, and z_{ij} is the monetary intermediate flow from i to j. The resulting value is an annual benchmark-flow-equivalent cost-incidence amount in current USD, not a cash loss accrued during seven calendar periods. For visualization, only international edges with $c(i) \neq c(j)$ are retained before extracting high-value links. This step unfolds the node results to the edge level; it does not add a new transmission mechanism.

The link matrix shows which transaction edges carry the accumulated cost burden, while topology analysis helps explain why some nodes repeatedly become relay points. To reduce noise from fragmented low-value dependencies, the analysis applies a common threshold κ to the technical-coefficient matrix and constructs the filtered directed backbone adjacency matrix $\mathcal{B} = (b_{ij})$ as

$$b_{ij} = \mathbb{I}\{\, a_{ij} \geq \kappa \,\}, \tag{19}$$

$$\mathrm{Indeg}_j = \frac{1}{n-1} \sum_i b_{ij}, \tag{20}$$

$$\lambda v_j = \sum_i b_{ij} v_i, \tag{21}$$

$$\mathrm{Bet}_i = \sum_{u \neq v \neq i} \frac{\sigma_{uv}(i)}{\sigma_{uv}}, \tag{22}$$

$$\mathrm{PR}_j = \frac{1-d}{n} + d \sum_i \frac{b_{ij}}{\sum_k b_{ik}} \mathrm{PR}_i. \tag{23}$$

In Equation (19), κ is the technical-coefficient threshold used to retain input dependencies in the backbone network. The baseline uses $\kappa = 0.001$, so a retained supplier accounts for at least 0.1% of the downstream node's gross output through a_{ij}. This cutoff is used only for topology interpretation; it is not applied to the propagation equations or to country, sector, link, and final-demand accounting. Supplementary Figure S1 reports robustness to moderate changes in κ. Equations (20)–(23) define normalized indegree, eigenvector centrality, betweenness, and PageRank. They capture breadth of upstream dependence, core placement, bridging position, and overall connection importance, respectively.

These topological indicators do not enter the propagation equations. They are used only to check whether the propagated burdens align with supply-chain position. Firm-level and network studies show that node size alone is not sufficient to explain systemic importance [39,40]; related spillover studies make the same point for production networks [41,42]. The study computes

$$\rho_q = \mathrm{Spearman}(q_i, C_i^{(H)}), \qquad q_i \in \{\mathrm{Indeg}_i, v_i, \mathrm{Bet}_i, \mathrm{PR}_i\}. \tag{24}$$

In Equation (24), ρ_q is the rank correlation between topological indicator q_i and subsequent cascade pressure $C_i^{(H)}$. These correlations are structural checks, not causal estimates.

2.6. Final-Demand Incidence and Resilience Threshold Surface

The cumulative node-level costs must ultimately be realized through consumption, investment, and government demand. The analysis maps the cumulative propagated vector $P^{(H)}$ onto the final-demand matrix F and obtains the final-demand cost matrix. $L^{(H)}$ identifies where propagated costs are finally absorbed in national demand rather than where the shock first enters the production network.

$$L^{(H)} = \mathrm{diag}(P^{(H)})F, \tag{25}$$

$$l_{ik}^{(H)} = P_i^{(H)} f_{ik}, \tag{26}$$

$$B_k^{(H)} = \sum_i l_{ik}^{(H)}, \tag{27}$$

where $l_{ik}^{(H)}$ is the amount of propagated cost generated by node i and ultimately absorbed by the final demand of destination country k, and $B_k^{(H)}$ is the total propagated final-demand cost burden of destination country k. As with the link matrix, monetary values are annual benchmark-flow-equivalent incidence measures based on the 2022 ICIO table. They do not measure the amount of money lost during a seven-round calendar interval. Direct source-node final sales are not added separately; the calculation follows the same $P^{(H)}$ convention used in the link matrix. If the result is further aggregated by source economy u, one obtains

$$Y_k^F = \sum_i f_{ik}, \qquad R_k^{(H)} = 100 \times \frac{B_k^{(H)}}{Y_k^F}. \tag{28}$$

Equation (28) defines the relative final-demand burden reported in Section 3.5. The denominator Y_k^F is nominal destination final demand from the cleaned production-by-final-demand block used to construct F: production rows are the cleaned ICIO production nodes, while tax, value-added, total, and non-production rows are excluded. The original OECD final-demand account categories are retained and reaggregated by destination country, including inventory-change entries where they appear in the ICIO final-demand block. The ratio is therefore a current-USD benchmark-flow incidence share, not a deflated real burden or an observed inflation rate.

If the result is further aggregated by source economy u, one obtains

$$B_{u \to k}^{(H)} = \sum_{i \in N_u} l_{ik}^{(H)}. \tag{29}$$

Equation (29) identifies the production countries or regions from which the final cost burden originates. It does not identify the exact route through which that burden travels. The stacked bars reported in the final-demand results separate domestic sources, Chinese production sources, and other foreign sources. This is a model-implied measure of final expenditure pressure, not an observed CPI component or an assumption of instantaneous retail pass-through. This interpretation follows work on inflation spillovers and systemically significant prices [43–45].

The inventory boundary cannot be identified from a single country ranking. The analysis reruns the propagation process on a grid of upstream-energy shock level α and inventory absorption rate γ, using the average added cost of global downstream manufacturing outside the initially shocked sectors as the system-state indicator. The grid varies the intensity of the same shock composition: $\alpha_U = 0.75\alpha$, $\alpha_T = 1.25\alpha$, and $\eta = 0.50\alpha$. Let $M \subset D$ denote the downstream manufacturing nodes, and let ω_i^M be the output-normalized weight of manufacturing node i. Then

$$\Phi(\alpha, \gamma; H) = \sum_{i \in M} \omega_i^M P_i^{(H)}(\alpha, \gamma), \tag{30}$$

$$\alpha^*(\gamma; \tau) = \inf\{\, \alpha : \Phi(\alpha, \gamma; H) \geq \tau \,\}, \qquad \tau = 10\%. \tag{31}$$

Equation (30) gives the weighted average added cost under a given shock and inventory setting. Equation (31) defines the minimum upstream-energy shock level required to reach threshold τ. The reference value $\tau = 10\%$ provides a common scale across scenarios; it is not a hard real-economy tipping point. The indicator excludes source sectors so that it reflects network transmission rather than the exogenous energy and transport disturbance itself.

3. Results

The results are interpreted as energy supply resilience indicators rather than as a general network-ranking exercise. The source block is deliberately composed of energy and energy-logistics accounts: oil and gas extraction, refining, electricity and gas utilities, water and land transport, warehousing, and transport-support services. These accounts determine whether fuel availability, delivered energy cost, and transport access remain stable enough for downstream production to continue. The later MRIO rounds then show which non-source sectors absorb the residual burden after inventories, reserves, and initial logistics adjustments have dampened part of the shock.

The baseline results are reported with a propagation window of $H = 7$ and with inventory damping turned on. The presentation follows the transmission chain itself. The analysis begins with country-level entry exposure and later-round burden, then turns to sectoral cost-transfer positions, high-value cross-border links, inventory-buffering limits, and final-demand incidence. Supplementary Tables S4–S6 report the leading country, sector, and cross-border link values behind the main figures in numeric form.

Table 4 makes the energy-system layer explicit before the broader propagation results are read. It reports the baseline shock values assigned to the initially affected fuel, refining, utility, transport, and logistics sectors and links them to the main routes through which costs reach downstream users. Energy sectors are therefore treated as the entry layer through which fuel availability, energy-service continuity, and delivered logistics costs first change.

Table 4. Energy-system entry accounts, baseline shock values, and policy-relevant pass-through routes.

Energy-System Account	Baseline Shock	Main Pass-Through Route	Mitigation Focus
Oil and gas extraction	20%; +10% for Gulf energy nodes	Resource-to-refining links carry the largest values, including ROW_B06 to CN1_C19 and ROW_B06 to IND_C19.	Substitute barrels; bypass capacity; reserve drawdown
Refining	20%; +10% for Gulf energy nodes where applicable	Crude availability and quality mismatch are converted into fuel and feedstock costs for chemicals, utilities, transport, and manufacturing.	Crude-slate matching; product swaps; refinery-run continuity
Utilities	15%	Electricity, gas, steam, and related energy services pass costs into metals, chemicals, port services, cold-chain users, and public services.	Fuel allocation for power and gas; critical-load protection
Transport	25%	Longer routes, tanker scarcity, freight escalation, and war-risk premia appear in maritime and fuel-intensive transport receivers.	Rerouting; vessel scheduling; energy-cargo priority
Logistics and transport support	25%	Port handling, warehousing, clearance, storage, and inland movement determine whether replacement cargoes reach refineries and industrial users quickly.	Terminal and storage priority; port clearance; inland logistics coordination

Note: ROW is the residual rest-of-world aggregate in the ICIO table. CN1 denotes the Chinese subregional node retained for link extraction before national reaggregation.

3.1. First-Round Exposure and Later-Round Burden Do Not Coincide

Figure 1 decomposes country-level risk into first-round direct exposure $E_c^{(1)}$ and subsequent network cascades $C_c^{(H)}$. In the calibrated sample, the two are positively related overall, with a Spearman correlation of about 0.78 and a linear fit with $R^2 = 0.57$. The relationship is meaningful, but it is not tight enough to treat entry exposure and cumulative burden as interchangeable measures. If direct Gulf dependence were a sufficient summary of vulnerability, the points in Figure 1 would cluster much more closely around the fitted line. The observed dispersion shows that this is not the case. Once the shock enters the network, later-round amplification changes relative country positions in a systematic way.

The largest deviations from the fitted relation show how much the MRIO pass-through calculation changes a direct-exposure ranking. Several Asian production economies have positive residuals: their first-round values are not always the largest, but their output-weighted downstream sectors keep receiving added cost in later rounds. The mechanism is a cost-accounting one. Sectors in these economies buy material, energy-service, and transport-service inputs that already contain upstream cost increases, so $C_c^{(H)}$ remains high after the entry round has passed. Ukraine appears at the top of the benchmark subsequent-burden table, but this should be treated as a structural outlier. The value reflects the weighted 2022 ICIO production structure, including the still-recorded energy-intensive industrial base, and does not reconstruct wartime operating conditions or the more recent integration of Ukraine into European energy and fuel-supply channels. We therefore retain Ukraine in the benchmark table for transparency but do not use it as the substantive basis for the country interpretation. Supplementary Table S4a reports the same ranking after excluding Ukraine; Vietnam, Bahrain, Malaysia, China, and Thailand then form the leading group. By contrast, the United States and several energy exporters face visible entry exposure while showing more limited later-round accumulation. Singapore is the clearest case in which the first-round value is tied to shipping, transshipment, and energy-trading functions.

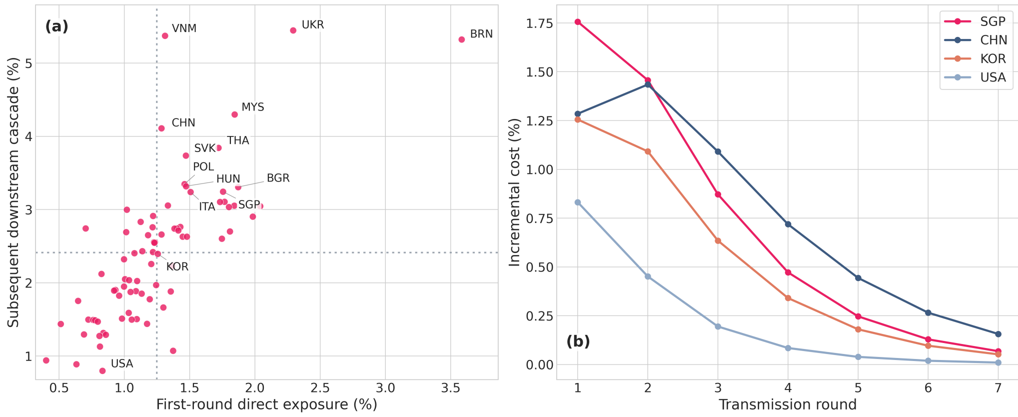

Figure 1. First-round direct exposure, subsequent network cascades, and round-by-round incremental costs at the country level. (**a**) Relationship between first-round exposure and subsequent cascades across economies. Grey dotted reference lines indicate the sample-average values of first-round direct exposure and subsequent network cascade pressure. (**b**) Incremental cost paths over seven transmission rounds for selected economies.

The round-specific paths show the same separation in time. Singapore records incremental costs of roughly 1.76% and 1.46% in rounds 1 and 2, followed by a rapid decline. China starts at about 1.28% in round 1, rises to about 1.43% in round 2, and only then begins to fall. Korea declines from 1.26% to 1.09%, while the United States drops from 0.83% to 0.45%. These paths show more than who is high or low. They identify which economies experience a short entry spike and which economies remain under pressure over several rounds. A country that peaks immediately and then retreats faces a different timing problem from a country whose burden persists between rounds 2 and 4.

Taken together, the evidence shows that entry exposure is an incomplete ranking device. Entry indicators are useful for locating the source of the shock, but the MRIO pass-through calculation is needed to rank downstream cost incidence. Under the blockade scenario, the difference between the two rankings widens because energy disruption and logistics disruption are imposed together. This pattern is consistent with earthquake and firm-network evidence on supplier–customer propagation [46,47], pandemic evidence on production-chain disruption [26–28], and input–output studies of cascading dependence [29,48,49].

3.2. Energy-System Entry Sectors and Later-Round Industrial Burdens

Country rankings show where added cost is absorbed. Sector rankings identify the accounts that carry those costs through the industrial structure. The initial source block is an energy-system block rather than a generic shock vector. Oil and gas extraction and refining represent the physical fuel-availability channel; utilities represent the energy-service channel through electricity, gas, steam, and related services; transport and transport-support sectors represent the delivered-cost channel through routing, insurance, port handling, warehousing, and inland movement. These sectors are the first policy concern because they determine whether reserve barrels, alternative cargoes, or substitute routes can actually enter production.

Figure 2 reports sector-level subsequent cascades for sectors outside the initial shock block. The top positions are C24A, H51, C20, C23, and C24B, followed by C22, C25, C27, motor vehicles, and machinery and transport equipment. In economic terms, these are material-processing, aviation-service, chemical, mineral, metal, and equipment-related accounts with large purchases from energy- and logistics-intensive suppliers. Once a transmitted cost increase enters these accounts, it is recorded in a broad set of downstream production costs rather than only in the source energy balance.

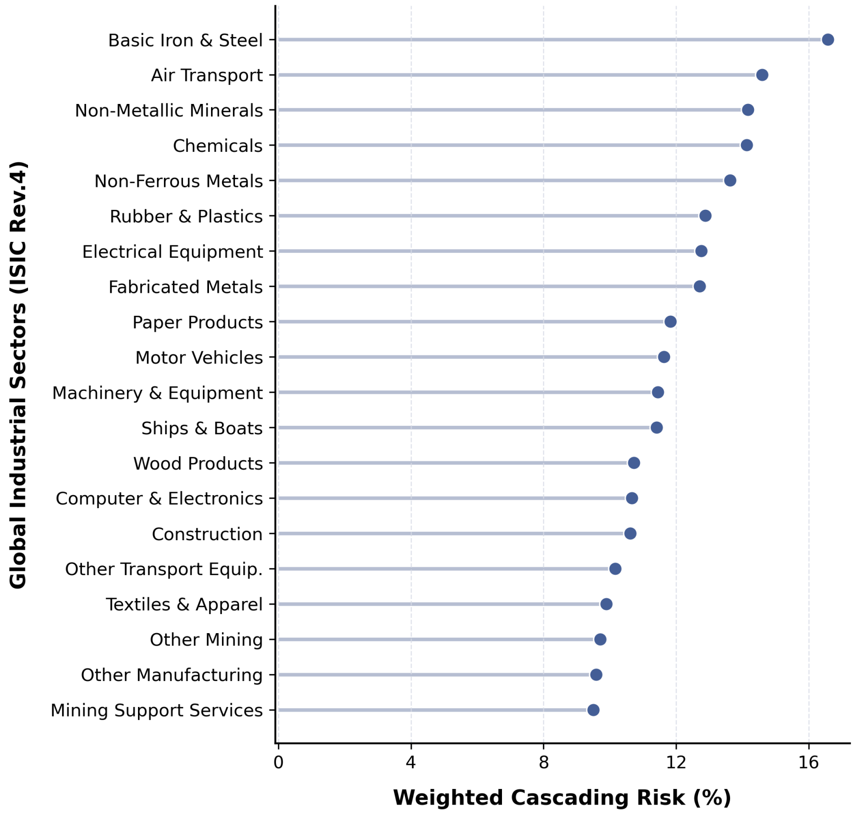

Figure 2. Sector-level subsequent cascades outside the initial shock block. The ranking excludes the initially shocked energy, utility, transport, and transport-support sectors so that later industrial burden is not mechanically dominated by source-sector shocks.

Air transport deserves explicit comment because it is outside the initial source block but still ranks high in the sectoral results. In the cost-transfer matrix, aviation-related services receive large propagated input-cost values and then sell into high-value manufacturing and service users. This is different from claiming that air transport is an exogenous shock source in the baseline vector. Utilities and selected transport-support sectors have a different status because they are part of the initial shock block, yet they also remain embedded in the transactions through which later-round pressure is redistributed.

Figure 3 extends this interpretation into the cross-sector dimension. Horizontal dark bands identify the main cost-transmitting sectors, while vertical dark bands identify the main cost-receiving sectors. The transmitting side is concentrated in utilities, refining, land and water transport, warehousing support, chemicals, and basic metals. The receiving side includes construction, wholesale and retail trade, computers and electronics, and food and beverages. The off-diagonal blocks indicate sectoral reallocation of added cost from source and material accounts to industrial users. The heatmap should be read as accounting pass-through across observed input purchases, not as evidence that one sector mechanically causes output failure in another.

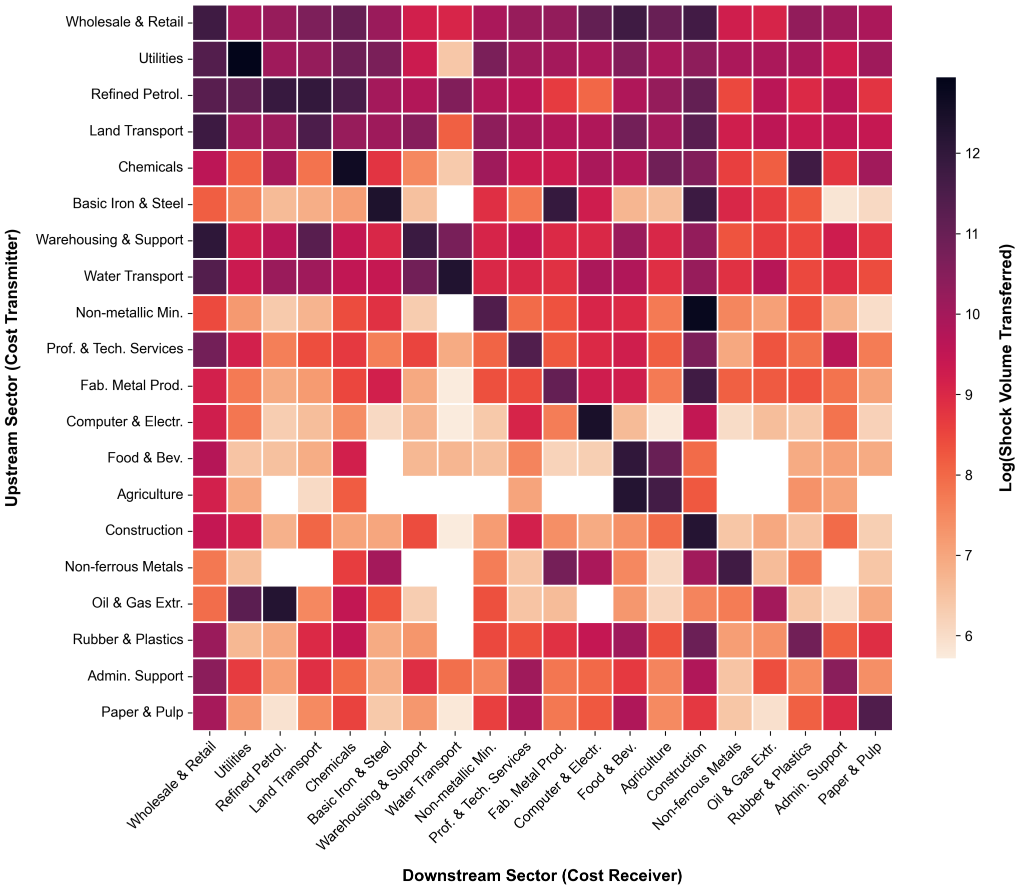

Figure 3. Cross-sector cost migration after seven cumulative rounds. Darker cells indicate larger cumulative transfers between transmitting and receiving sectors in the observed ICIO input-purchase structure.

The sector results also narrow the interpretation of source-side intervention. Expanding crude supply or refining capacity reduces the entry wedge, but the downstream cost account is also shaped by material-processing and logistics-service purchases already embedded in production. Entry sectors remain important. The additional policy problem is how to prevent high-value intermediate-input links from converting a temporary energy-logistics premium into a wider manufacturing cost burden.

3.3. High-Value Cross-Border Links and Their Structural Interpretation

The sector matrix shows where pressure lingers. The upper panel of Figure 4 identifies the international transaction edges through which that pressure continues to move. After seven cumulative rounds, the major propagated downstream transfers fall into four broad groups. The first links energy extraction to refining sectors in the United States, China, India, and Korea. The second consists of intra-transport maritime links. The third connects electronics sectors across the East Asian production system, including Chinese subregional nodes, Taiwan, Vietnam, and Korea. The fourth links Australian minerals to metal-material processing in China. These edges show that amplification is carried by a limited number of high-value channels, not by a diffuse spread across all international transactions.

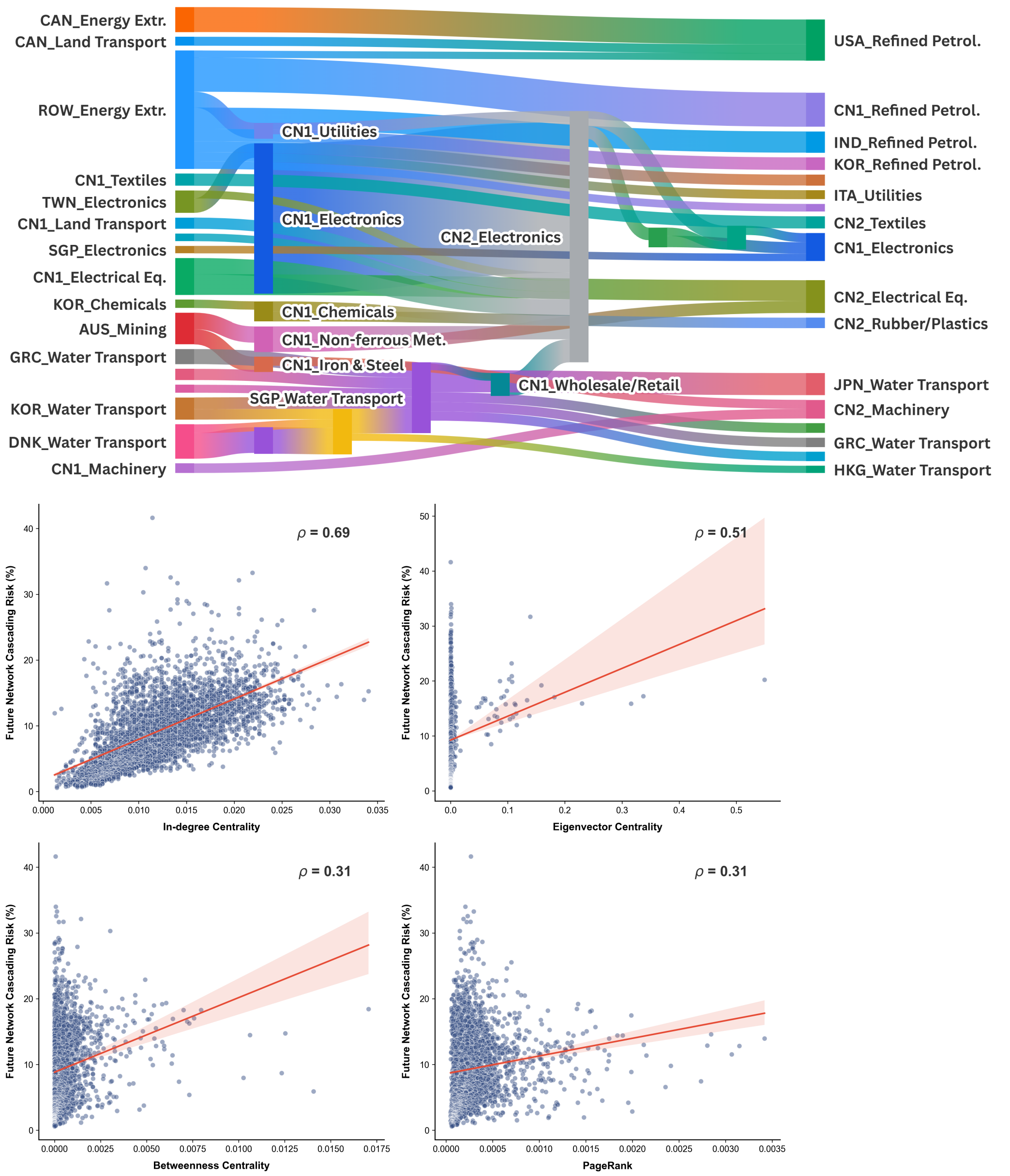

Figure 4. High-value cross-border cost links and their topological correlates. The upper panel reports the main cross-regional high-value links after seven cumulative rounds; colored blocks denote country-sector accounts, and band width is proportional to the propagated cost-transfer value. In the lower panel, blue dots are country-sector observations, red lines are fitted linear trends, and pale red bands are the corresponding confidence intervals. The lower panel reports the Spearman correlations between topological metrics and subsequent cascade pressure. CN1 and CN2 denote the original Chinese subregional nodes retained for link extraction before national reaggregation; ROW denotes the residual rest-of-world aggregate.

The four groups also point to distinct accounting channels. Resource-to-refining edges attach the transmitted energy-cost increase to refining and primary energy-conversion activities. Maritime links record cost stacking and redistribution within the transport system itself. Electronics links in East Asia show how transport and material-cost increases are recorded in assembly-related accounts. Resource-to-material edges show the role of metals and mineral processing as interfaces between energy costs and physical manufacturing inputs. The link results show that absolute transfer value, not geographic proximity to the strait, explains why some destination sectors keep receiving cost pressure.

The largest links in Supplementary Table S6 make this concentration visible in numeric form. The leading transfers are from ROW energy extraction (B06) to China refining (CN1_C19; USD 4.608 billion), from ROW energy extraction (B06) to India refining (IND_C19; USD 2.870 billion), from Canadian energy extraction (CAN_B06) to U.S. refining (USA_C19; USD 2.227 billion), from ROW energy extraction (B06) to China utilities (CN1_D; USD 2.141 billion), and from China water transport (CN1_H50) to Singapore water transport (SGP_H50; USD 1.778 billion). ROW is the residual rest-of-world aggregate in the ICIO table, not a single economy; CN1 is the Chinese subregional node retained for link extraction. These values identify the transactions that should be monitored first if the policy objective is to prevent an energy-route shock from becoming a broader refining, utility, and logistics bottleneck.

The lower panel of Figure 4 provides a structural interpretation of why the same links recur in the propagation backbone. The Spearman correlation between indegree centrality and subsequent cascade pressure is about 0.69. The corresponding value for eigenvector centrality is about 0.51, while betweenness and PageRank are both around 0.31. This ranking suggests that the breadth of upstream dependence and core placement in the supply chain matter more than bridging position alone. Nodes with many upstream inputs can receive cost pressure from several directions at once. Nodes embedded in the core can then pass that pressure into a wider range of downstream activities. The topological evidence is consistent with the sector results. Basic materials, transport-support activities, and complex manufacturing accounts remain under sustained burden because they combine large transaction scale with broad sourcing depth and strong embedding in the production core.

These topological measures are used only for structural interpretation. They do not enter the propagation equations, and the analysis does not claim that a specific centrality index causes cascades. They show that the sectors and cross-border links highlighted by the propagation results are located in supply-chain positions that are already more likely to accumulate and retransmit cost pressure.

3.4. Effectiveness Boundary of Inventory Buffering

In the cost-pressure model, inventory damping rescales onward pass-through but leaves the dominant MRIO transfer paths largely unchanged. The left panel of Figure 5 compares country-level subsequent cascade pressure under scenarios with and without inventory. Korea shows the largest reduction in subsequent burden, followed by Hungary, Slovakia, and Vietnam. Korea is in the baseline buffer pool, while Hungary, Slovakia, and Vietnam benefit indirectly because inventory damping in upstream supplier economies reduces the residual passed further downstream. The high-burden group remains similar after damping is introduced. More generally, the inventory term lowers the level of measured added cost without replacing the input–output paths that carry it.

The right panel of Figure 5 writes that result as a threshold problem. The 10% downstream manufacturing added-cost contour reports the combinations of upstream-energy shock level and inventory absorption at which the average added cost of global downstream manufacturing outside the initially shocked sectors reaches 10%. When the inventory absorption rate is zero, the critical upstream-energy shock level required to reach that threshold is about 73.6%. As the inventory absorption rate rises to 10%, 20%, 30%, 40%, and 50%, the critical value increases to 77.7%, 81.6%, 85.3%, 88.9%, and 92.2%, respectively. Under the 20% baseline upstream-energy shock, the average added cost of global downstream manufacturing falls from 2.72% to 2.34%, a reduction of about 0.38 percentage points, or roughly 14% in relative terms. Inventories matter in a measurable way, but their effect remains bounded.

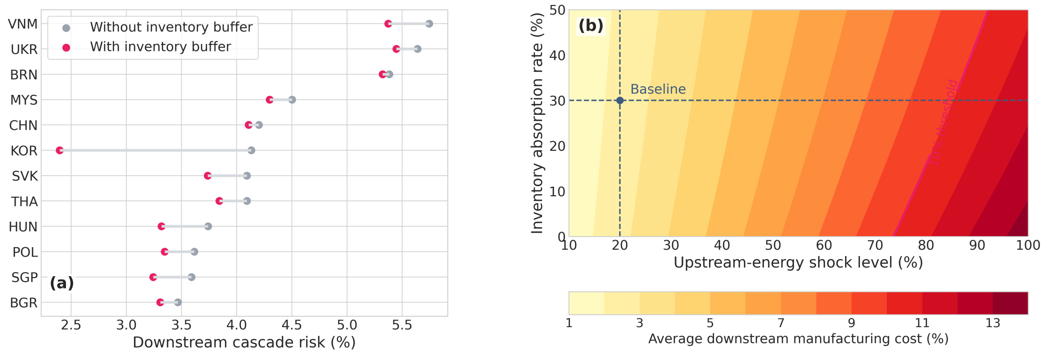

Figure 5. The bounded effectiveness of inventory buffering and the threshold boundary. (**a**) Country-level subsequent cascade pressure with and without inventories. (**b**) The 10% downstream manufacturing added-cost boundary on the two-dimensional grid of upstream-energy shock level and inventory absorption rate. For Ukraine, the reported position mainly reflects the structure of the weighted 2022 benchmark table rather than a short-run operational forecast.

The threshold increments narrow as the buffer rises. A 10-percentage-point increase in the inventory absorption rate raises the critical upstream-energy shock level by 4.1 percentage points when γ moves from 0 to 10%, then by 3.9, 3.7, 3.6, and 3.3 percentage points across the remaining grid intervals. Moving from low to moderate inventory levels shifts the boundary more visibly; further increases still help, but by less. The damping parameter lowers the measured added-cost level at each point on the grid, while the ranking of high-burden country-sector accounts remains stable.

Two stronger inventory checks support this interpretation. When China is added to the buffer-country pool, China's subsequent cascade burden falls from 4.11% to 2.11%, but the rank correlation with the baseline country ranking remains 0.973. When the baseline buffer pool is retained but damping is made sector-specific, the average downstream manufacturing added cost under the 20% upstream-energy shock rises from 2.34% to 2.46%, and the 10% boundary falls from 85.3% to 81.4%. This is more conservative in the sense that it relaxes the broad economy-wide buffering assumption: only energy, utility, and logistics nodes in buffer economies retain the 0.30 damping value, while selected bottleneck sectors receive 0.20 and other sectors receive 0.10. More residual pressure is therefore passed forward than under the common 0.30 rule. The country ranking remains close to the baseline, with a rank correlation of 0.992. These checks suggest that inventories reduce the amount passed on; they do not replace the main input–output routes through which the shock travels.

Inventories can widen the distance between the system and the 10% downstream manufacturing added-cost threshold under medium-intensity upstream-energy shocks and can extend the adjustment window for substitute sourcing, shipping adjustment, and production rescheduling. They reduce onward pass-through rather than rerouting the cost-transfer paths, so the inventory boundary is best interpreted together with the country, sector, and link results.

Table 5 summarizes the critical upstream-energy shock levels corresponding to the inventory absorption rates reported in the threshold grid.

Table 5. Critical upstream-energy shock level for a 10% downstream manufacturing added-cost threshold under alternative inventory absorption rates.

Inventory Absorption Rate γ	Critical Upstream-Energy Shock Level α^* (%)
0%	73.6%
10%	77.7%
20%	81.6%
30%	85.3%
40%	88.9%
50%	92.2%

Note: α^* is the upstream-energy component of a jointly scaled energy-logistics stress vector. At each grid point, the utilities shock, transport/logistics shock, and Gulf-node premium are set to 0.75α, 1.25α, and 0.50α, respectively. The values are threshold indicators, not forecasts of oil prices, freight rates, or macroeconomic losses.

3.5. Where Cumulative Costs Are Ultimately Absorbed Across Final Markets

Figure 6 performs a final allocation step. It multiplies the propagated node-cost vector by the destination-country final-demand matrix and reallocates intermediate cost pressure to expenditure destinations. In the calibrated results, the largest absolute entries are China, the United States, the rest-of-world aggregate, Germany, Japan, India, Italy, and France. The values are reported in current USD billion under the annual ICIO benchmark-flow-equivalent convention. They represent model-implied propagated final-demand incidence, not source-inclusive total losses, observed price indices, or losses accrued over seven calendar periods.

China absorbs about USD 614.1 billion in total, of which about USD 598.8 billion comes from domestic sources and about USD 15.3 billion from foreign sources. China's large domestic-source component means that cost pressure first transmitted through intermediate inputs is then assigned to Chinese final demand through Chinese production nodes; it should not be read as a direct foreign import bill. The United States absorbs about USD 259.5 billion in total, including about USD 199.4 billion from domestic sources, about USD 15.3 billion from Chinese production sources, and about USD 44.8 billion from other foreign sources.

Normalizing these annual benchmark-flow-equivalent values by destination final demand changes the scale of interpretation. The denominator is the nominal destination final demand Y_k^F defined from the same cleaned production-by-final-demand block used in Equation (28), not an external GDP or value-added series. The same calculation equals about 3.69% of final demand for China, 2.81% for Italy, 1.98% for the ROW aggregate, 1.75% for India, 1.71% for France, 1.65% for Germany, 1.46% for Japan, and 1.00% for the United States. The United States therefore ranks second in absolute burden but lower after normalization because its final-demand base is larger. China remains high under both measures because the modeled burden is large relative to its destination final demand as well as large in absolute terms.

The rest-of-world aggregate reported in the figure is the residual ICIO destination group rather than a single economy. Chinese production sources refer to the Chinese production nodes after the original ICIO subregional nodes are reaggregated for country-level reporting. The ordering differs from the ranking of first-round direct exposure because Figure 6 weights propagated node costs by destination-country expenditure.

Table 6 reports the corresponding absolute and normalized burden values for the leading final-demand destinations.

The composition of the burden is as informative as its total. For the major destination accounts, domestic absorption remains the largest component, yet external pressure from Chinese production sources and from other foreign sources is also substantial. Germany absorbs about USD 17.1 billion from foreign sources outside China, while India absorbs about USD 2.2 billion from Chinese production sources. These values show how upstream and intermediate cost entries are converted into consumption, investment, or government-demand incidence once the final-demand matrix is applied.

Table 6. Absolute and normalized final-demand burden for the leading destinations.

Destination	Burden (USD Billion)	Share of Final Demand (%)
China	614.1	3.69
United States	259.5	1.00
Rest of world (ROW)	91.0	1.98
Germany (GER)	61.9	1.65
Japan	61.6	1.46
India	57.3	1.75
Italy	54.7	2.81
France	46.4	1.71

Note: ROW denotes the residual rest-of-world aggregate in the ICIO table, not a single economy. GER is used here as the common abbreviation for Germany; the underlying ICIO code is DEU.

Final-demand incidence is a model-based destination of cumulative cost pressure, not a direct forecast of CPI components or retail-price adjustment. The calculation identifies where propagated costs are assigned after production accounts are connected to destination expenditure. For that reason, the destination ranking can differ from the first-round exposure ranking even when the underlying source shock is unchanged.

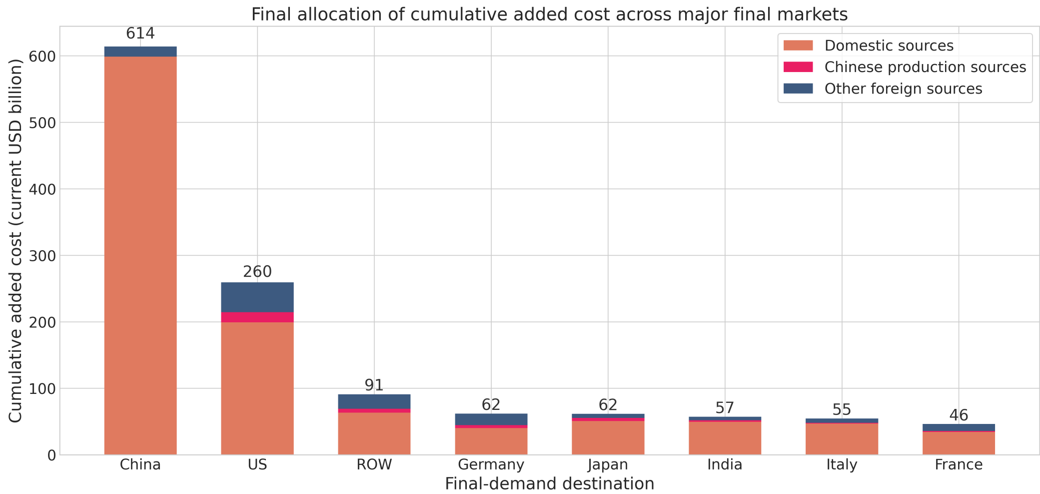

Figure 6. Destination of propagated cost burdens on the final-demand side. The bar chart aggregates cumulative costs by destination country, reports annual benchmark-flow-equivalent values in current USD billion, and separates domestic sources, Chinese production sources, and other foreign sources. ROW denotes the residual ICIO rest-of-world destination.

4. Policy Implications for Energy Supply Resilience

The results frame a Strait of Hormuz shock as an energy supply resilience and production-continuity problem, not as an oil-price shock alone. Recent Red Sea disruptions show that longer routes and tighter shipping schedules can absorb effective transport capacity even when aggregate trade volume does not collapse [50]. Energy-agency reports on Hormuz underline why a blockade would add a more direct energy constraint on top of that logistics strain [1–3]. The policy implications are organized around the points where intervention can reduce pass-through. Reserve release acts first on fuel availability and refining continuity. Routing and logistics coordination act on the delivery of replacement energy cargoes and critical inputs. Inventory management buys time for substitute sourcing and production rescheduling. Industrial-continuity measures protect the downstream sectors that convert energy and logistics pressure into wider production burdens. These policy implications should be read as intervention points implied by the stress test, not as optimized emergency-response prescriptions. The model does not optimize the timing or size of reserve releases; it identifies the production accounts and transaction links where reserve release, substitute routing, logistics coordination, and inventory management are most likely to reduce downstream pass-through. Table 7 links each result to the corresponding policy lever and monitoring indicator.

Table 7. Result-to-policy matrix for a Hormuz energy-route disruption

Result from the Stress Test	Policy Objective	Instrument	Lead Actor	Trigger Indicator	Expected Effect
Oil, gas, and refining nodes receive the initial cost increase	Maintain fuel and refinery continuity	Strategic petroleum reserve release; product swaps; crude-slate matching	Energy ministries; reserve agencies; refiners	Crude import shortfall; refinery run cuts; product inventory drawdown	Lower the initial cost increase before it reaches utilities and manufacturing
Transport and logistics-support sectors transmit delivered-cost pressure	Keep replacement cargoes and critical inputs moving	Rerouting; tanker scheduling; port-slot priority; customs fast lanes	Port authorities; maritime agencies; customs; carriers	War-risk premium; freight rate; port dwell time; vessel arrival delay	Limit modeled logistics pass-through and cost stacking in transport services
High-value cross-border links concentrate residual burden	Protect the transactions that carry the largest propagated costs	Dual sourcing; backup contracts; near-port storage; priority handling for listed links	Trade agencies; firms; port operators	Disruption on top resource-to-refining, maritime-service, electronics, and minerals-to-metals links	Limit conversion of an energy-route shock into broader industrial cost pressure
Inventory damping lowers pass-through but does not change main routes	Buy time for substitute sourcing and production rescheduling	Coordinated release of public reserves, commercial inventories, and firm stocks	Governments; industry associations; large firms	Days of cover for crude, products, feedstocks, and critical intermediates	Delay the 10% downstream manufacturing added-cost boundary
Final-demand incidence differs from first-round exposure	Stabilize markets where expenditure pressure accumulates	Working-capital support; trade finance; procurement adjustment; targeted consumer or utility support where justified	Finance ministries; development banks; procurement agencies	Final-demand burden share; producer prices; import prices; household energy bills	Prevent propagated production costs from becoming avoidable final-market stress

4.1. Reserve Release, Routing, and Logistics Coordination at the Entry Stage

The country and link results indicate that the entry stage should not be framed only as a reserve question. Reserves matter, but the binding constraint can also appear in the links that connect reserves and imports to the productive system. A strategic petroleum reserve release reduces effective pressure only if the released crude or products match refinery configurations, reach the right terminals, and move through storage, pipelines, ports, and inland transport quickly enough. For LNG, spare cargoes, destination flexibility, regasification slots, and gas-grid constraints play a similar role. When crude, LNG, chemicals, or metal feedstocks arrive late, fail to clear ports quickly, or cannot be moved onward to refineries and industrial users, the effective shock is larger than the physical supply loss alone.

Route substitution should be treated with the same operational caution. Saudi and UAE bypass routes, product swaps, alternative tanker schedules, and cargo diversions can reduce the immediate supply gap, but they do not automatically remove war-risk premia, vessel shortages, port queues, or inland distribution bottlenecks. Reserve release, substitute procurement, port clearance, hinterland transport, and utilities support therefore need to be coordinated around a short list of country-sector links connecting import entry points to refining, electricity and gas services, basic materials, and export-oriented manufacturing. Those links are where a temporary interruption is most likely to become a broader production shock.

For manufacturing economies with high output-weighted downstream burdens, the refinery–utilities–materials chain and the port–warehouse–factory chain need to remain connected. That requires unloading and storage capacity for crude and LNG, transport services between ports and refining or heavy-industry clusters, emergency clearance rules for a narrow list of critical inputs, and coordination with insurers, ship operators, port authorities, customs agencies, and inland carriers. If these entry links remain open, later-round transmission is moderated where intervention remains most direct. If they fail, the burden appears in material, logistics-service, and manufacturing accounts rather than staying at the import terminal.

4.2. Inventory Management and Industrial-Continuity Protection

The inventory results point to a bounded but still important role for stockholding. Inventories delay the approach to the 10% downstream manufacturing added-cost threshold, while the sectors and links through which pressure accumulates remain broadly similar. They are best used to buy time around the high-value input links identified in Section 3, not treated as a sufficient policy instrument on their own. Public reserves are best suited to stabilizing refining, electricity, gas, and basic transport services because these sectors transmit cost pressure quickly into the rest of the system. Commercial inventories are more useful for petrochemical feedstocks, metals, and other intermediate materials that sit between energy and manufacturing. Firm inventories matter most where order fulfillment depends on short delivery windows, especially in electronics, machinery, and equipment production.

Industrial-continuity policy should therefore be narrower than economy-wide support. The priority sectors are not only the largest energy users; they are the sectors that link energy availability to downstream production schedules. Refining, electricity and gas utilities, port and warehousing services, basic chemicals, metals, air transport, electronics, machinery, and transport equipment need contingency rules for fuel allocation, emergency procurement, overtime logistics capacity, and temporary credit support. These measures do not remove the shock. They reduce the probability that a temporary energy-route interruption becomes a stoppage in sectors with large downstream purchasing relationships.

The timing of inventory release affects what inventories can actually accomplish. Reserve release should be evaluated by whether it reduces pass-through into the highest-value material and logistics input links. Once drawdown begins, governments and firms need an explicit transition from stock release to substitute sourcing, process adjustment, and logistics reprioritization. Otherwise the initial benefit is only delayed, not preserved. This division of roles is inferred from the cost-transfer results; it is not an optimization result from the model. Monitoring should follow the same logic. National import dependence is too coarse for operational use. The relevant indicators include war-risk premia, freight rates, port dwell time, and vessel arrival delays [50,51]. They also include refinery and chemical operating rates, days of inventory for critical intermediates, and delivery schedules for major industrial parks and export clusters [52–54]. These are the metrics that reveal whether the system is moving toward the threshold boundary identified in Figure 5.

4.3. Final-Market Stabilization and Selective Redundancy

The final-demand results add a third policy layer. Policy aimed at final markets should be guided by destination-side cost incidence, since domestic supply chains can convert upstream cost pressure into expenditure pressure before retail prices are observed. Recovery planning therefore cannot stop at import security and continuity of high-value input links. For large manufacturing exporters, the relevant instruments include trade finance, working capital, and contract execution capacity among downstream firms once higher input costs begin to compress margins. For large domestic markets, the relevant monitoring problem is where cost transmission is most likely to spill into household demand, investment programs, and procurement budgets. These implications follow from the burden-distribution results; they are not direct quantity estimates. The model does not forecast CPI directly, but it does identify where final expenditure pressure is likely to accumulate.

The same results support targeted redundancy before broad reshoring. The edges highlighted in Figure 4 show that a small number of high-value cross-border channels carry a disproportionate share of the cumulative burden. Those are the links that justify supply diversification through dual sourcing, regional backup capacity, alternative shipping plans, and near-port or near-plant buffer storage for critical inputs. A blanket effort to relocate whole sectors would spread policy resources too widely and would do little to protect the specific input nodes that govern cascade intensity. A narrower strategy is more feasible and better aligned with the results. It focuses on the links that combine high transaction value, strong network embeddedness, and repeated appearance in later-round propagation. That is where structural resilience can be improved without discarding the wider gains from international specialization.

5. Limitations

The results should be interpreted within the data and indicator boundaries of the exercise. The empirical base is the 2022 OECD ICIO benchmark, which keeps the transaction matrix, final-demand accounts, and sector definitions internally consistent but cannot reconstruct later wartime structural changes in Ukraine or other post-2022 country-specific adjustments. Ukraine's high subsequent-burden ranking is therefore treated as a benchmark-structure outlier rather than as an operational forecast for 2026. The reported country and sector burdens are also production-cost incidence indicators. They are weighted by gross output because the propagation equations and technical coefficients are defined per unit of gross output; they should not be read as value-added losses, GDP losses, or welfare losses. This choice gives visibility to intermediate-intensive sectors, but it can overweight activities with large intermediate purchases relative to sectors with high value added.

The stress test also uses a deliberately short-run price and technology structure. Benchmark unit prices are normalized to one, and relative cost wedges are applied to the fixed 2022 input structure. This identifies where additional cost pressure would be recorded if input shares were held constant, but it does not estimate equilibrium price adjustment, demand contraction, exchange-rate responses, expectations, or firm-level switching behavior. Holding input shares fixed is equivalent to assuming zero substitution elasticity among intermediate inputs during the modeled pass-through process. That assumption is defensible for an immediate crisis window because contracts, technical specifications, certification requirements, shipping schedules, refinery configurations, and supplier qualification procedures cannot be reorganized instantly. It becomes stronger in later propagation rounds. The seven rounds used here are accounting transmission rounds rather than seven calendar periods, but later-round results should still be read as exposure embedded in the benchmark procurement structure before adaptive substitution is explicitly modeled, not as realized costs after firms have had sufficient time to redesign supply chains and production recipes.

The scenario assumptions are kept simple for comparability, which also narrows their interpretation. Common utility and transport shocks make the cross-country experiment transparent, but they may overstate domestic-source burdens for large diversified economies whose electricity systems, fuel mix, and domestic transport are less directly tied to Gulf oil and gas imports, including China and the United States. Scaling these shocks by country-specific dependence on Gulf oil and gas would likely reduce the domestic-source component for diversified systems and concentrate it more among highly import-dependent economies. Inventory damping is likewise reduced-form. China is excluded from the baseline buffer pool because the baseline rule is restricted to directly comparable commercial-inventory information and explicit U.S. strategic petroleum reserve information; this is a data-consistency decision rather than a claim that China lacks emergency stocks or industrial inventories. Supplementary Table S3 shows that adding China to the buffer pool lowers China's subsequent cascade burden from 4.11% to 2.11% and raises the downstream manufacturing threshold beyond the tested upstream-energy shock range. That result is useful as a sensitivity check, but it also shows that the global and East Asian distribution of cascading risk can change materially when a large economy is assigned the same inventory-damping rule as the baseline buffer economies. For this reason, the expanded China-pool case is not used as the baseline, and the baseline results should be read as conditional on the stated inventory-buffer rule. Future work should link country-specific reserve data, commercial inventories, sectoral stockholding, release feasibility, routing, and logistics data to the damping and shock parameters before using such scenarios for operational country ranking.

6. Conclusions

This study examines how a severe Strait of Hormuz disruption would propagate through global production accounts when energy supply, route access, and freight costs are shocked together. Using the 2025 OECD ICIO release for 2022, the paper builds a dynamic MRIO stress test in which oil and gas extraction, refining, utilities, transport, and transport-support activities form the initial shock block. The results show that first-round exposure and cumulative burden do not coincide. Later-round costs are shaped by the output-weighted purchase of energy-, material-, and logistics-intensive inputs, and high-value transmission is concentrated in a limited set of resource-to-refining, maritime-service, electronics, and minerals-to-metals links.

Inventories reduce the amount passed on, but they do not replace the main transmission routes. When the inventory absorption rate rises from 0 to 50%, the upstream-energy shock level associated with the 10% downstream manufacturing added-cost threshold increases from 73.6% to 92.2%. Under the 20% baseline upstream-energy shock, the average added cost of global downstream manufacturing outside the source sectors declines from 2.72% to 2.34%. The final-demand calculation adds a destination-side result: propagated costs are ultimately assigned to expenditure markets, so the final-demand ranking differs from the first-round exposure ranking.

For energy-system policy, the main implication is that Hormuz chokepoint risk should be managed as a supply-resilience and industrial-continuity problem, not only as an oil-price or import-dependence problem. Reserve release, substitute routing, port and freight coordination, inventory management, and protection of critical industrial links need to be considered together because the shock enters through fuel, refining, utilities, transport, and logistics accounts before reaching manufacturing and final demand. Future work should combine route- and port-specific logistics data with more explicit stock-release, substitution, and firm-adjustment mechanisms.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/en19112719/s1, Parameter anchoring for the baseline Strait of Hormuz scenario, shock-source sector mapping, robustness checks for the propagation window and topology filter, stronger inventory-damping robustness scenarios, and numeric verification tables for the main figures. S1. Energy-Route Scenario Anchoring and Shock Mapping: Table S1. Parameter anchoring for the baseline Strait of Hormuz energy-route stress scenario. S2. Energy-System Shock-Source Sectors and Gulf Export Nodes: Figure S1. Robustness checks for the propagation window and the topology threshold; Table S2. Energy-system shock-source sector mapping and Gulf export node set. S3. Robustness Checks and Boundary Tests. S4. Stronger Inventory-Damping Checks: Table S3. Stronger inventory-damping robustness scenarios. S5. Numeric Verification Tables: Table S4. Top countries by subsequent cascade burden; Table S4a. Country ranking after excluding Ukraine as a benchmark-structure outlier; Table S5. Top sectors by subsequent cascade burden; Table S6. Top cross-border cost links after seven cumulative rounds.

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